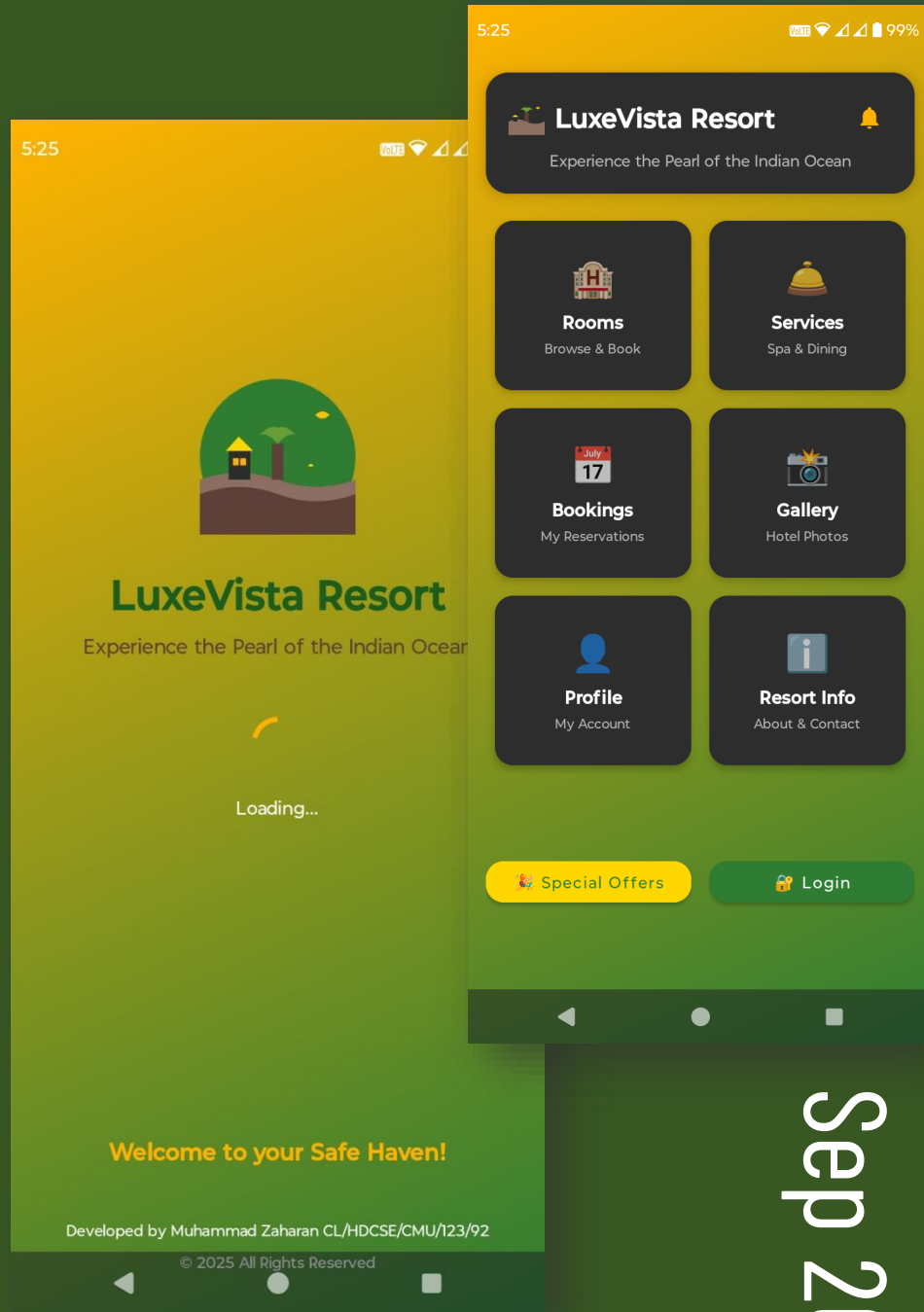


The LuxeVista Resort Mobile Application System

Developed By Muhammad Zaharan
CL/HDCSE/CMU/123/92



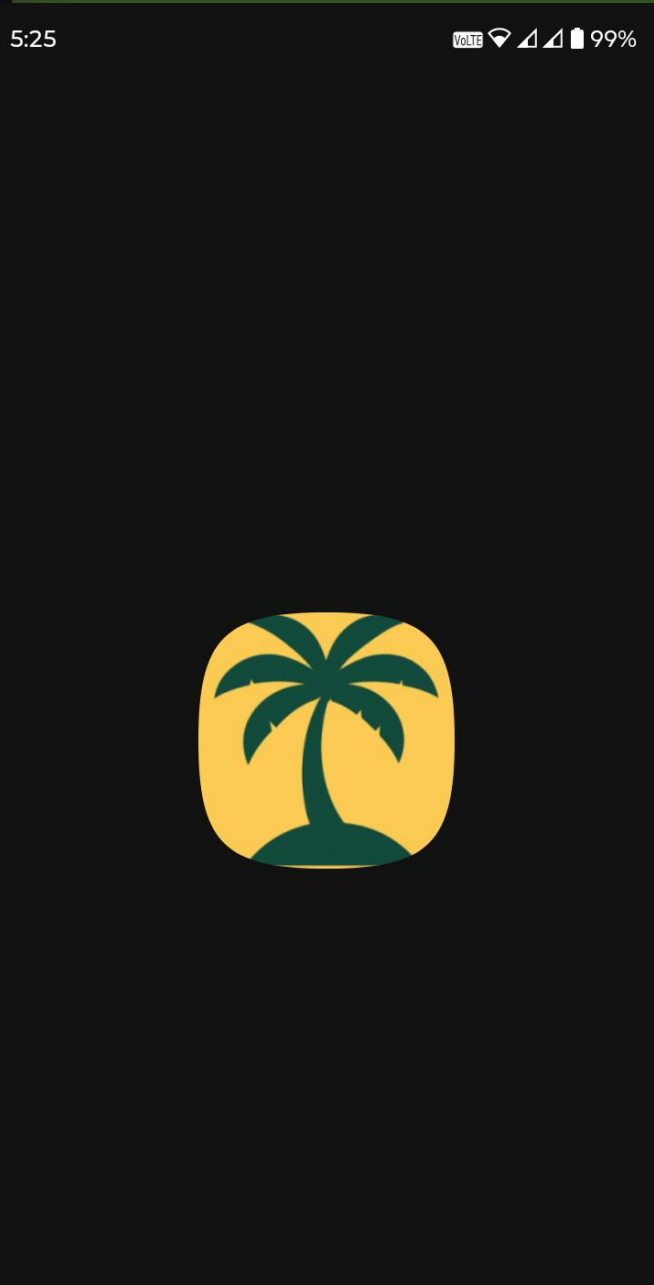
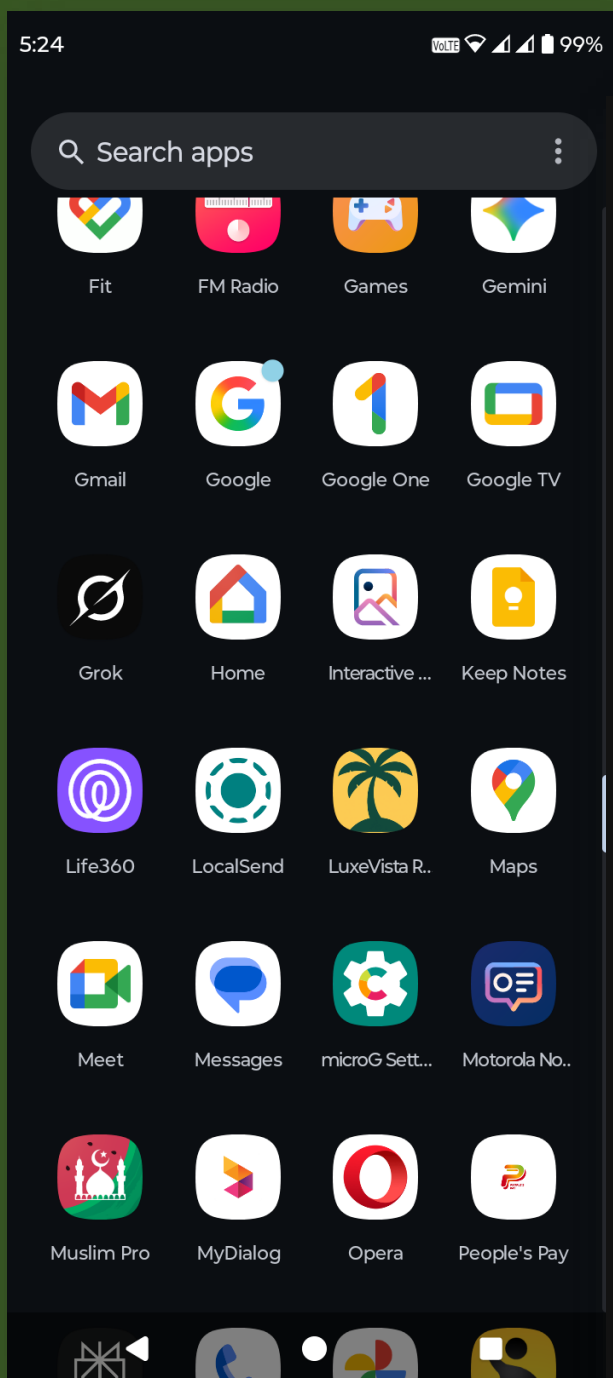
Sep 2025

1. Introduction

The LuxeVista Resort mobile application provides guests with a seamless digital concierge to book rooms, reserve services, and manage their stay.

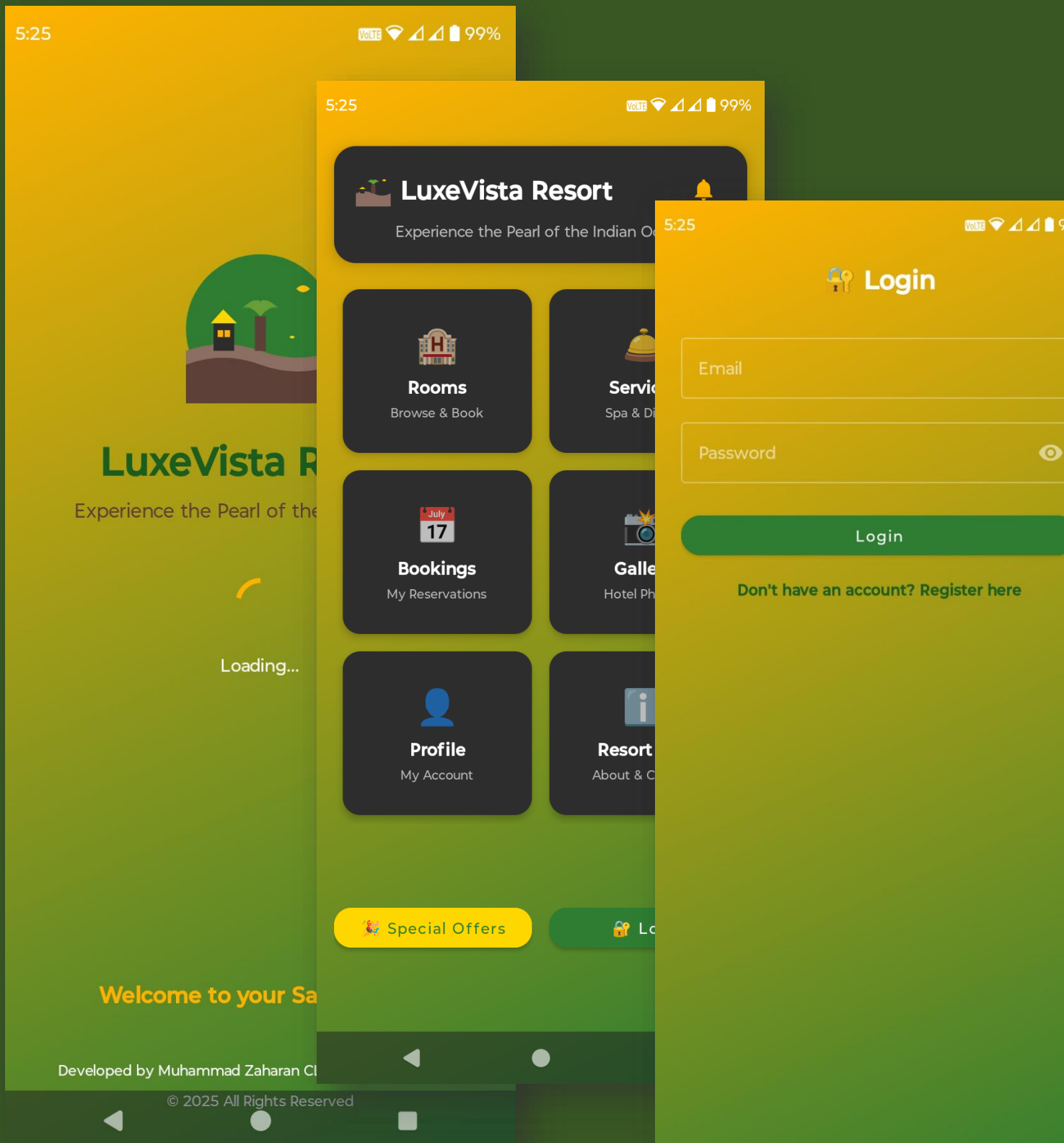
2. Installation

- Available as an Android APK.
- Download from the official link or install via the provided QR code.
- Requires Android 8.0 (Oreo) or later.



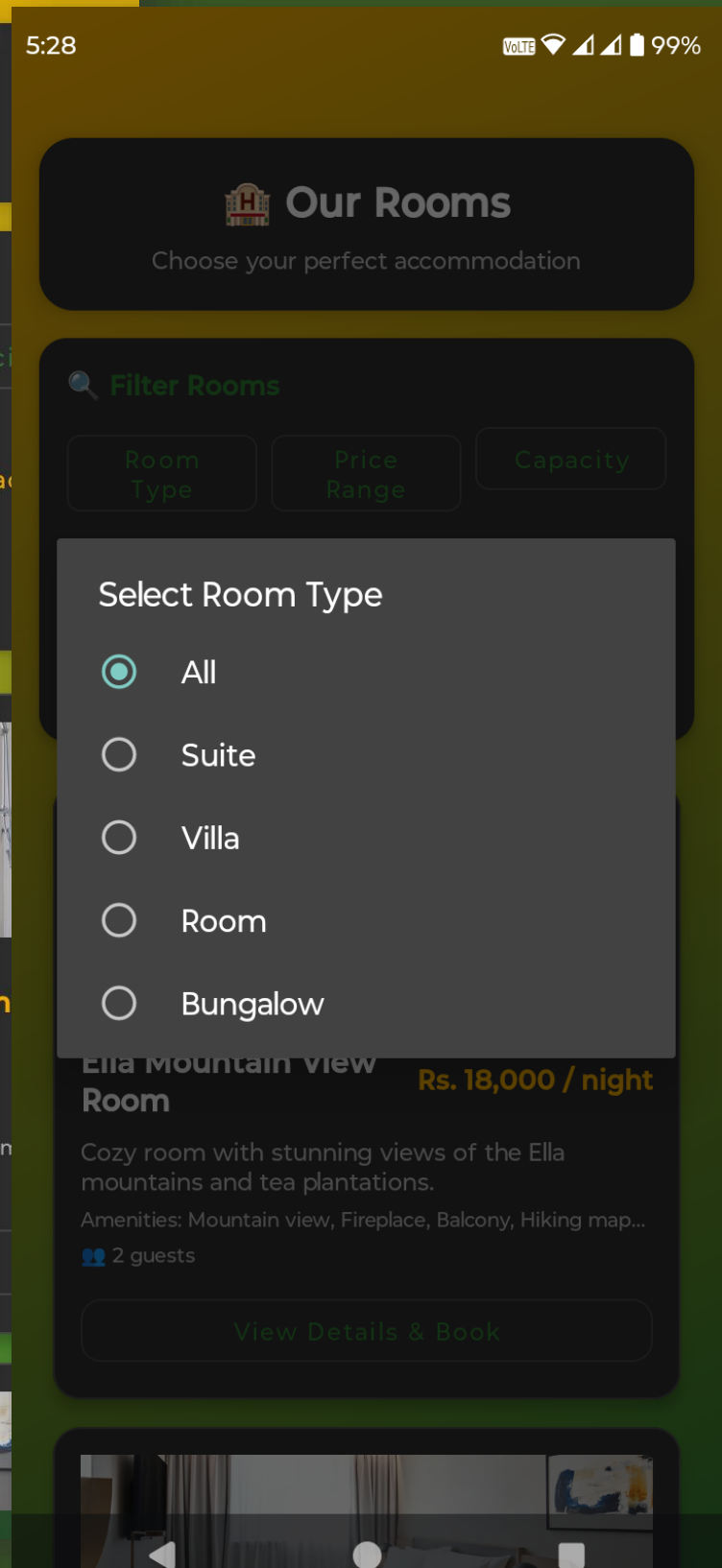
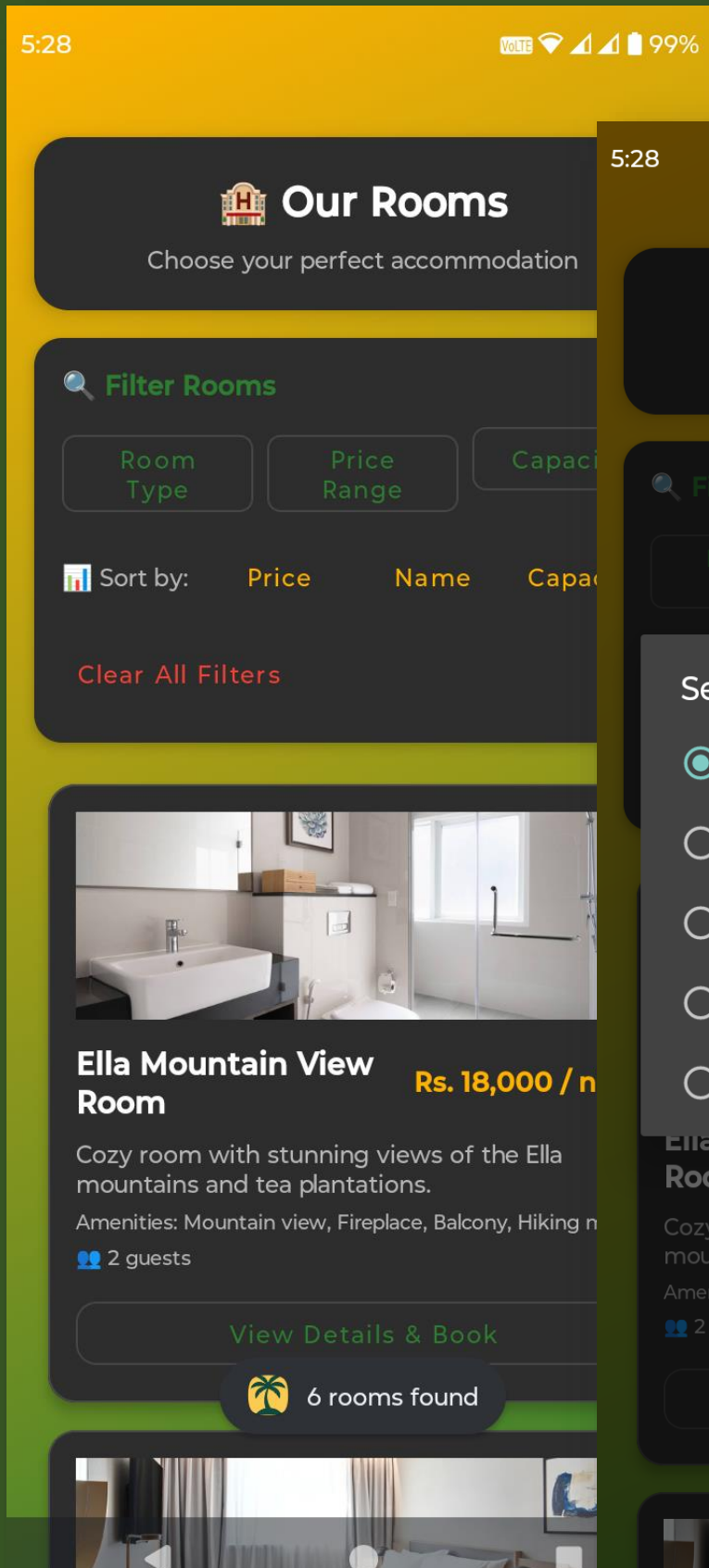
3. Getting Started

1. Launch the app – LuxeVista Resort splash screen is displayed.
2. Registration – new guests can create an account with name, email, and password.
3. Login – returning guests enter credentials to access their dashboard.



4. Features & Instructions

- Browse Rooms – view available rooms with descriptions, prices, and images.



4. Features & Instructions

- Book Rooms – select dates and confirm reservations.

5:28

VOLTE

99%

 **Our Rooms**
Choose your perfect accommodation

 **Filter Rooms**

Room Type

Price Range

Capacity

 Sort by:

Price ↓

Name

Capacity

Clear All Filters



Presidential Suite **Rs. 75,000 / night**

The ultimate luxury experience with butler service and exclusive amenities.
Amenities: City view, Butler service, Jacuzzi, Premium toil...
 2 guests

View Details & Book

 6 rooms found



5:28

VOLTE

99%

Ella Mountain View Room
Rs. 18,000.00 / night



Description

Cozy room with stunning views of the Ella mountains and tea plantations.

Amenities

Mountain view, Fireplace, Balcony, Hiking maps, Tea plantation views
 2 guests

Select Dates

Check-in: 2025-09-21

Check-out: 2025-09-22

Book This Room

4. Features & Instructions

- Reserve Services – choose from spa, dining, cabanas, and more with calendar-based scheduling.

5:29

VOLTE

99%

 **Resort Services**
Enhance your stay with our premium services

 **Filter Services**

Category

Price Range

Duration

 Sort by: **Price** **Name** **Duration**

Clear All Filters



Concierge Service **Rs. 2,500**
Personalized concierge service for all your resort and travel needs in Sri Lanka.
 As needed  24/7
Category: Concierge

Book Service

 Found 6 services



5:29

VOLTE

99%

Concierge Service
Rs. 2,500.00


Description
Personalized concierge service for all your resort and travel needs in Sri Lanka.

Service Details
 24/7
 Duration: As needed
Category: Concierge

Select Date & Time

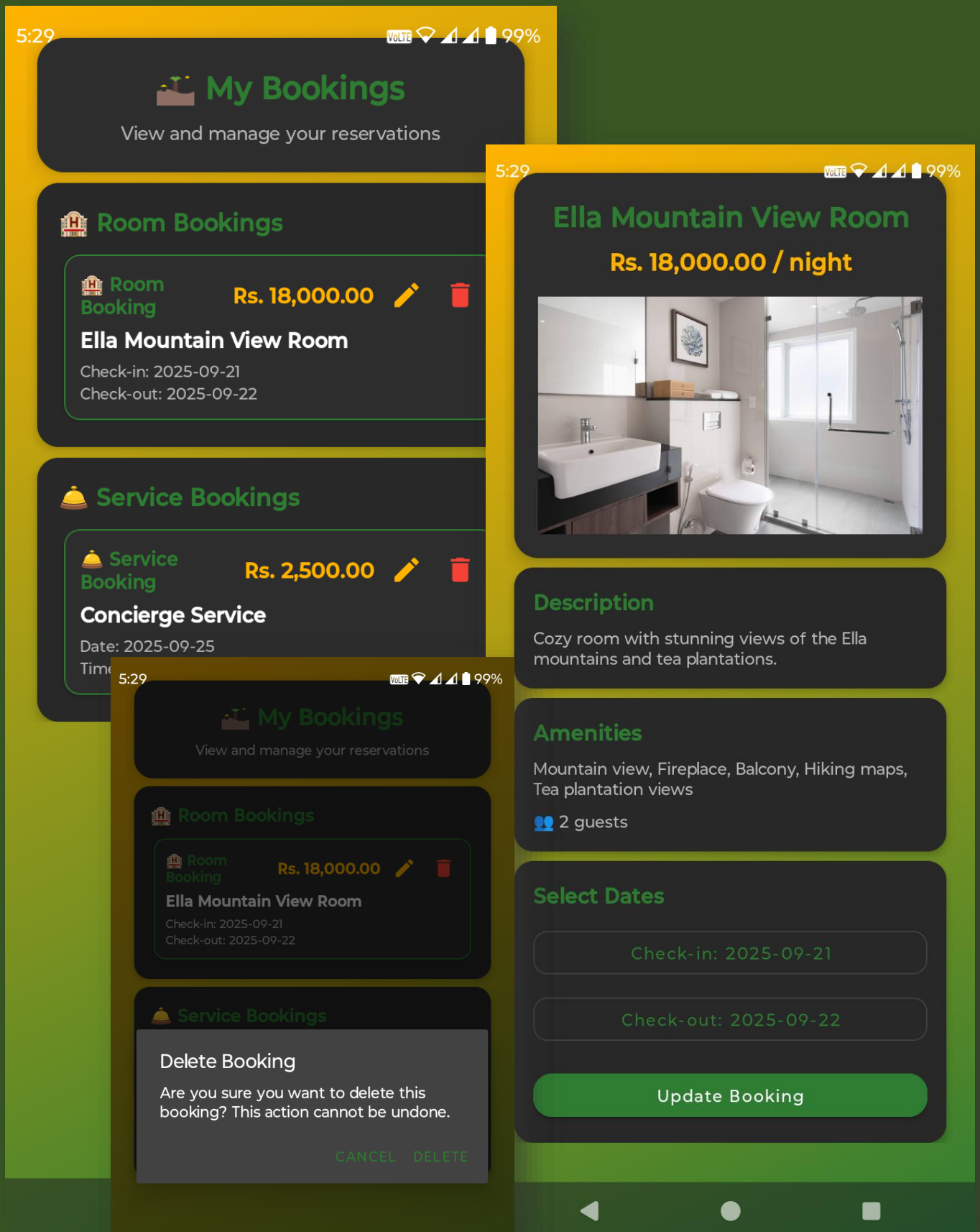
Date: 2025-09-20

Time: 06:00

Book This Service

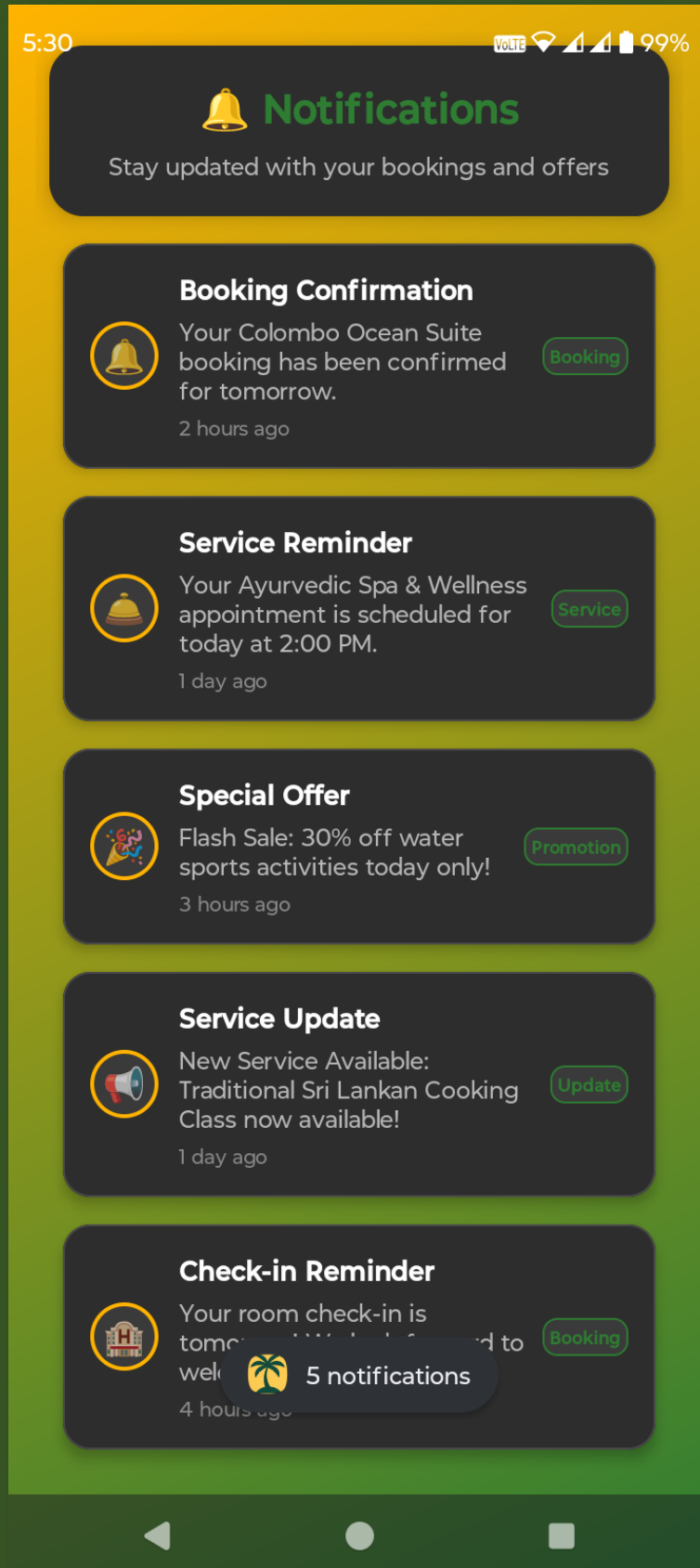
4. Features & Instructions

- Manage Bookings – review upcoming reservations in the “Bookings” section.



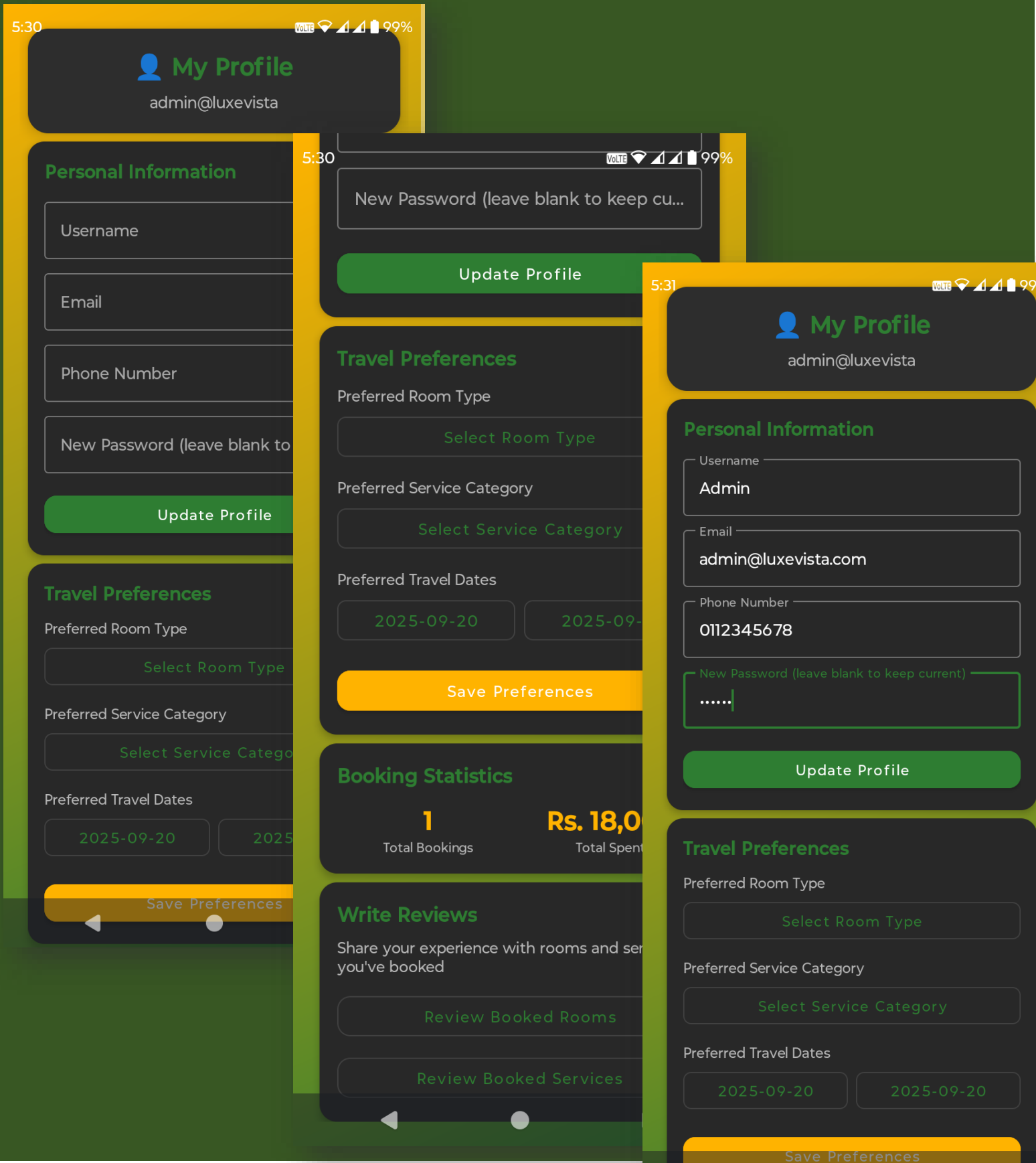
4. Features & Instructions

- Notifications – receive reminders about reservations and special promotions.



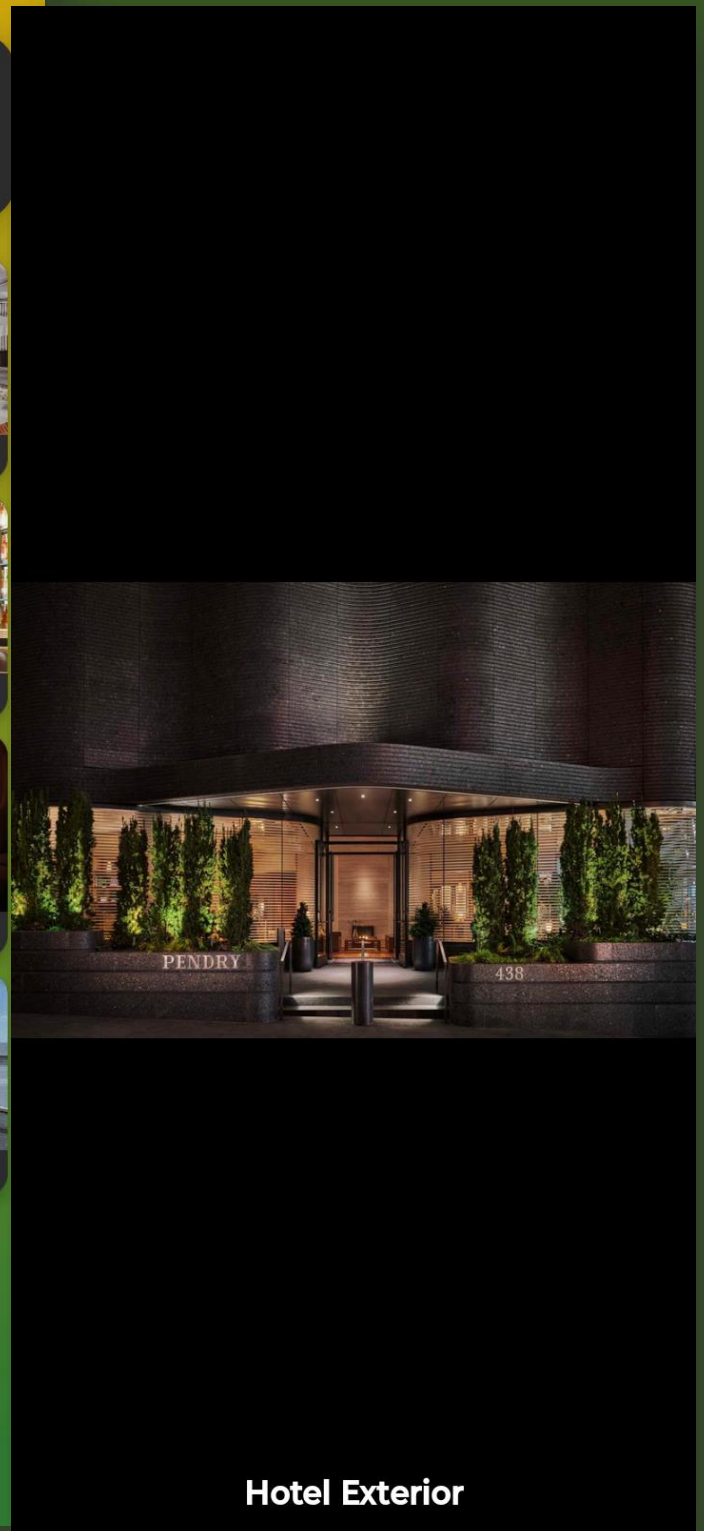
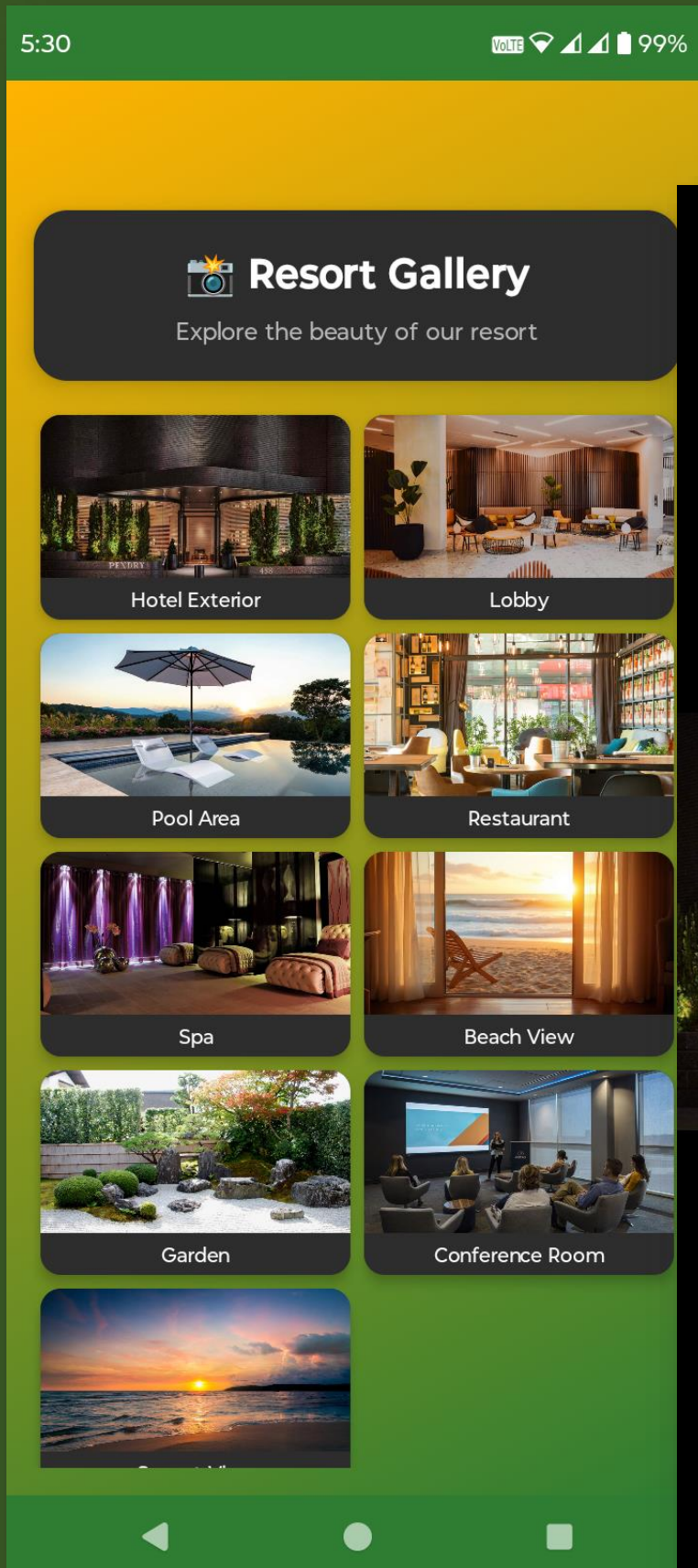
4. Features & Instructions

- Profile Management – update personal information and preferences



4. Features & Instructions

- Gallery -Displays uploaded hotel and facilities photos.



5. Help & Support

- For issues, contact LuxeVista Resort reception or app support.
- In-app notifications provide updates on maintenance and promotions.

5:32

VOLTE

99%

 **Write Review**

Ella Mountain View Room

Your Rating



Excellent

Write your review

Excellent service *100*

Submit Review

5:32

VOLTE

98%

 **Resort Information**

Everything you need to know about LuxeVista

About LuxeVista Resort

Welcome to LuxeVista Resort, where luxury meets the beauty of Sri Lanka. Nestled on the pristine shores of the Indian Ocean, our resort offers an unparalleled experience of relaxation and adventure in the Pearl of the Indian Ocean.

With world-class amenities, authentic Sri Lankan cuisine, and breathtaking ocean views, we provide the perfect escape for discerning travelers seeking the ultimate in comfort and service while experiencing the rich culture and natural beauty of Sri Lanka.

Sri Lankan Attractions

 Unawatuna Beach - 5 min walk

 Hikkaduwa Coral Reef - 10 min boat ride

 Galle Fort Lighthouse - 15 min drive

 Royal Botanical Gardens - On-site

 Water Sports Center - On-site

 Kandy Cultural Show - 2 hours drive

 Adam's Peak Hiking - 3 hours drive

 Pettah Local Markets - 30 min drive

 Pinnawala Elephant Orphanage - 1 hour drive

 Sigiriya Rock Fortress - 4 hours drive

Contact Us

 Colombo, Sri Lanka

 +94 11 234 5678

 info@luxevistaresort.lk

 www.luxevistaresort.lk

 24/7 Concierge Service

 Colombo International Airport - 30 min

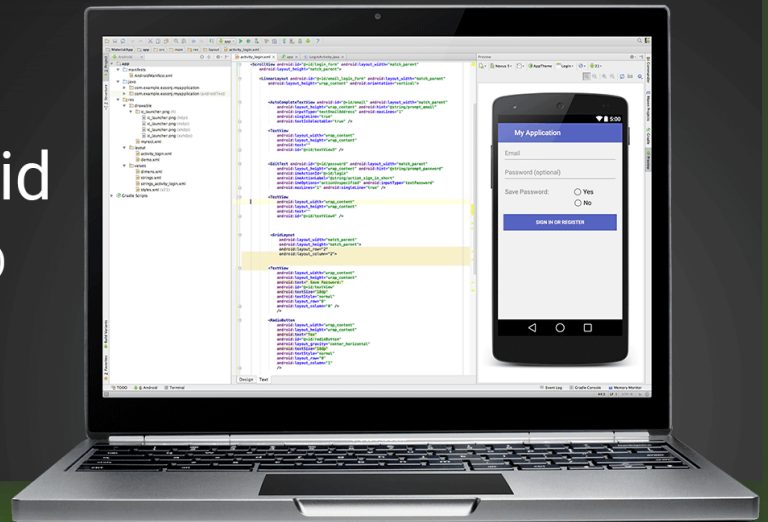
TECHNICAL DOCUMENTATION

Overview

- Platform: Android (Native)
- Language: Java
- IDE: Android Studio



Android
Studio



- Database: SQLite (local, embedded)
- Min SDK: API 26 (Android 8.0)



Architecture

- Activities: LoginActivity, RegisterActivity, RoomsActivity, ServicesActivity, BookingsActivity, ProfileActivity, RoomDetailActivity, ServiceDetailActivity.
- Adapters: RecyclerView adapters for rooms, services, and bookings.
- Database: DatabaseHelper for schema creation and CRUD operations.
- Session Management: SessionManager class for login persistence.
- Notifications: NotificationService for reminders.



Database Schema (Simplified)

users(id, name, email, password_hash, role)

rooms(id, name, description, price, capacity, amenities, image)

services(id, name, description, price, category, availability, duration)

bookings(id, user_id, room_id, service_id, booking_date, booking_time, status)

Constraints:

- UNIQUE(email) on users
- CHECK(price > 0, capacity >= 1)
- Foreign keys link bookings with users/rooms/services
- Unique indexes prevent double-booking

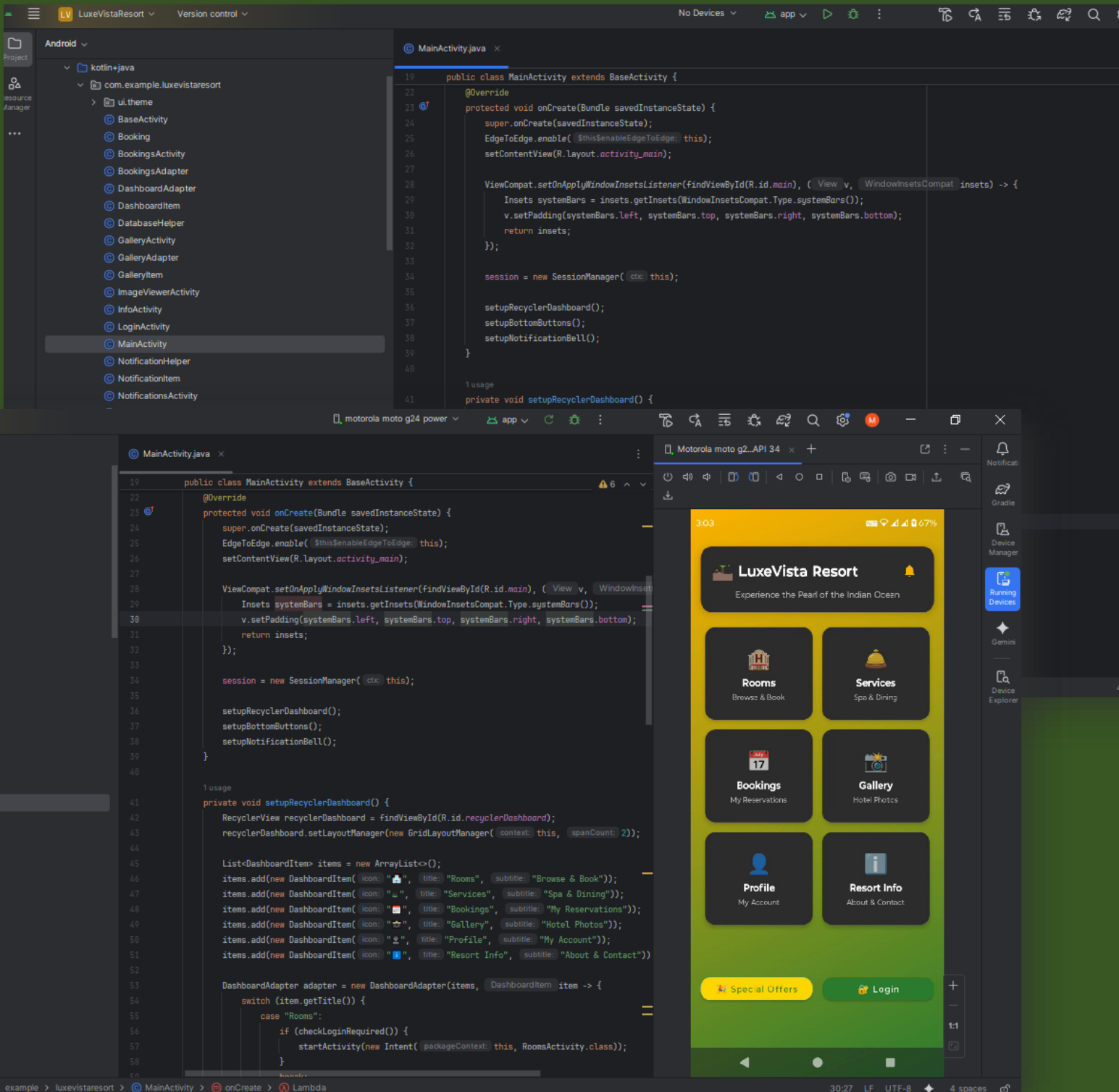
```
for (int i = 0; i < reviewerNames.length; i++) {
    ContentValues reviewValues = new ContentValues();
    reviewValues.put("user_id", 1); // Sample user ID
    reviewValues.put("item_id", itemIds[i]);
    reviewValues.put("review_type", reviewTypes[i]);
    reviewValues.put("reviewer_name", reviewerNames[i]);
    reviewValues.put("review_text", reviewTexts[i]);
    reviewValues.put("rating", ratings[i]);
    reviewValues.put("review_date", reviewDates[i]);
    // ... (insert logic) ...
}

public long addServiceBooking(long userId, long serviceId, String bookingDate, String bookingTime, double totalPrice) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("user_id", userId);
    values.put("service_id", serviceId);
    values.put("booking_date", bookingDate);
    values.put("booking_time", bookingTime);
    values.put("total_", totalPrice);
    return db.insert("bookings", null, values);
}

for (int i = 0; i < serviceNames.length; i++) {
    ContentValues serviceValues = new ContentValues();
    serviceValues.put("name", serviceNames[i]);
    serviceValues.put("description", serviceDescriptions[i]);
    serviceValues.put("price", servicePrices[i]);
    serviceValues.put("availability", serviceAvailability[i]);
    serviceValues.put("duration", serviceDuration[i]);
    serviceValues.put("category", serviceCategories[i]);
    serviceValues.put("image", serviceImages[i]);
    db.insert("services", null, serviceValues);
}
```

Setup Instructions

1. Open project in Android Studio.
2. Sync Gradle dependencies.
3. Run app on emulator or physical device.
4. On first run, database tables and seed data are created automatically.



Validation Mechanisms

Maintenance Notes